Takanao Tanaka

UC Berkeley
Agricultural and Resource Economics

Email: takanao_tanaka@berkeley.edu
Phone: +1-925(768)9343
Homepage: http://takanao-tanaka.com

Education

2021 –	PhD in Agricultural and Resource Economics, UC Berkeley
2019 – 2021	MPhil in Social Science, Hong Kong University of Science and Technology
2014 – 2016	M.A. in Public Policy, U Tokyo & National U of Singapore
2009 – 2014	B.A. in Economics, Keio University

Field Interest

Energy and Environmental Economics, Public Economics, Industrial Organization

References

Joseph Shapiro
Agricultural & Resource Economics
joseph.shapiro@berkeley.edu
(510) 859-4551

James Sallee
Agricultural & Resource Economics
sallee@berkeley.edu
(510) 643-5519

Reed Walker
Haas School of Business
rwalker@berkeley.edu
(510) 965-3298

Job Market Paper

Electricity Supply Shocks and Manufacturing: Evidence from the Fukushima Nuclear Accident (with Daiji Kawaguchi)

Abstract: We show that adverse shocks to electricity supply—an input accounting for only 2% of revenue—can cause substantial and persistent declines in industrial production. Using confidential establishment-level data and instrumenting with initial supply conditions, we find that shocks following the 2011 Fukushima nuclear accident reduced Japanese manufacturing electricity use by 6.6% and output by 1.7%, implying annual output losses of 4.9 trillion Japanese yen (50 billion US dollars). Establishments adjusted little through energy substitution, product switching, outsourcing, within-firm reallocation, or relocation. Our production function highlights two mechanisms that amplify output losses: establishments cannot easily substitute other inputs for electricity, and reduced electricity use leaves machinery and equipment temporarily idle. We quantify how electricity-market reforms, such as regional market integration, can mitigate these losses.

Working Papers

Combating Noise Pollution: Policy, Exposure, and Welfare (with Takeru Sugasawa and Shinsaku Takikawa)

Abstract: This paper studies how national noise-control policies affect noise exposure and welfare. We assemble two decades of largely untapped nationwide measurements of road-traffic noise from over 14,000 monitoring points in Japan and detailed data on noise-mitigation policies obtained through freedom-of-information requests. From 2002 to 2022, roadside noise fell by 2.8 decibels, equivalent to roughly a 20% decline in perceived loudness. We attribute about 70% of this decline to noise-control policies, particularly tighter vehicle noise-emission standards and the expansion of low-noise pavement. Residents are willing to pay 1.3% of residential land value for a one-decibel reduction in road-traffic noise. We estimate that Japan's national noise-control policies generated approximately 5 trillion yen (US$38 billion) in benefits over two decades. These benefits suggest substantial unrealized gains in countries with weaker noise-control policies, such as the United States.

Publications

1. Air Pollution and the Airborne Diseases: Evidence from China and Japan (with Guojun He and Yuhang Pan) ***Journal of Environmental Economics and Management***, 2025
2. Energy Saving May Kill: Evidence from the Fukushima Nuclear Accident (with Guojun He) ***American Economic Journal: Applied Economics***, 2023
3. Short and Medium-Term Impacts of Strict Anti-Contagion Policies on Non-COVID-19 Mortality in China (with Qi J, D Zhang, et al.) ***Nature Human Behaviour***, 2022
4. Increase in suicide following an initial decline during the COVID-19 pandemic in Japan (with Shohei Okamoto) ***Nature Human Behaviour***, 2021
5. The Short-term Impacts of COVID-19 Lockdown on Urban Air Pollution in China (with Guojun He and Yuhang Pan) ***Nature Sustainability***, 2020

Works in Progress

1. Environmental Regulation, Cost, Price and Welfare (*Census project approved*)

Fellowships, Awards, and Grants

2026 – 2028	Environmental Research Grant, Sumitomo Foundation, approx. $31,600.
2024 – 2026	Project Research Program Grant, IER Hitotsubashi University, $7000.
2022 – 2025	Japanese Student Services Organization (JASSO) Graduate Student Fellowship.
2022	17th Applied Econometrics Conference, Best Paper Award.
2020	15th Applied Econometrics Conference, Excellent Paper Award
2020 – 2021	2021 Annual iHEA Student Paper Prize, Second Prize.
2020 – 2021	Japan Center for Economic Research, Co-investigator, $9000.
2019 – 2021	Scholarship from Bai Xian Asia Institute.
2015 – 2016	Scholarship from Li Ka Shing Foundation.

Presentations

2026–27: ASSA Annual Meeting (D.C.); NBER Japan Project Meeting; Camp Resources; UC Berkeley (EERE)

2025–26: Institutional Barriers to Climate Adaptation (Stanford); LSE Environmental Camp (LSE); IPWSD (Columbia); UC Berkeley (EERE)

2021–24: AERE Annual Conference; JADE-JTREE Workshop (GRIPS); Osaka; SWET; Tsukuba; Occasional Workshop (UCSB); UC Berkeley (EERE)

Services

Referee Service: *American Economic Journal: Economic Policy, Journal of Environmental Economics and Management, Journal of Health Economics, Journal of Economic Behavior & Organization, China Economic Review, Economic Inquiry, Economic Modelling, Nature Climate Change, Scientific Reports*

Workshop Organizing: Student Organizer, EREE Workshop, UC Berkeley, 2024–25

Teaching Experience

UC Berkeley: GSIs for Environmental Economics, Intermediate Microeconomics (Undergrad)
HKUST: TAs for Research Methods in Social Science, Environmental Economics (Undergrad)

Professional Employment

2016 – 2018 Consultant, CSR Unit, Deloitte Consulting, Tokyo, Japan.